

PH 601: Mesoscopic Physics and Quantum Transport

Course Code	PH 601
Course Name	Mesoscopic Physics and Quantum Transport
L-T-P-C	3-0-0-3
Prerequisites	PH 501 Introductory Solid State Physics or teachers consent
Intended for	UG/PG Distribution: Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **Introduction** [5 Lectures]
- Drude and Sommerfeld model for electrons in solids, Quantum mechanics of particle in a box, Bloch states, Density of states and Dimensionality.
- **Mesoscopic physics** [5 Lectures]
- Mesoscopic phenomena and length scaling in physics, Quantum structures, Tunneling through the potential barrier, Coulomb blockade.
- **Quantum transport and Localization** [7 Lectures]
- Influence of reduced dimensionality on electron transport: Ballistic and Diffusive Transport, Single channel Landauer formula, Landauer-Buttiker formalism, Localization, Thermal activated conduction, Thouless picture, General and special cases of localization, Weak localization regime.
- **Quantum Hall effect** [8 Lectures]
- Origin of zero resistance, Two Dimensional Electron Gas, Transport in Graphene and two dimensional systems, Localizations in weak and strong magnetic fields, Quantum Hall effect, Spin Hall Effect.
- **Quantum interference effects in electronic transport** [6 Lectures]
- Conductance in mesoscopic systems, Shubnikov de Haas-Van and Aharonov-Bohm Oscillations, Conductance fluctuations.
- **Mesoscopic Physics with Superconductivity** [5 Lectures]
- Superconducting ring and thin wires, weakly coupled superconductors, Josephson effects, Andreev Reflections, Superconductor-Normal and Superconductor-Normal-Superconductor junctions.
- **Application of mesoscopic physics** [4 Lectures]
- Optoelectronics, Spintronics and Nanoelectronic Devices.

Text Books

1. Y. Imri, **Introduction to Mesoscopic Physics**, Oxford University Press, 2008.
2. S. Datta, **Electronic Transport in Mesoscopic Systems**, Cambridge University Press, 1997

References

1. S. Datta, **Quantum Transport: Atom to transistor**, Cambridge University Press, 2005.
2. B.L. Altshuler, P.A. Lee, R.A. Webb, (Editors), **Mesoscopic Phenomena in Solids** (Modern Problems in Condensed Matter Sciences), North Holland, 1991.
3. D. K. Ferry, S. M. Goodnick, **Transport in Nanostructures**, Cambridge University Press, 2009.
4. N. W. Ashcroft and N. D. Mermin, **Solid State Physics**, Cengage Learning, 1976.
5. P. Harrison, **Quantum Wells, Wires & Dots, Theoretical and Computational Physics of Semiconductor Nanostructures**, 2nd Edition, Wiley Science, 2009.